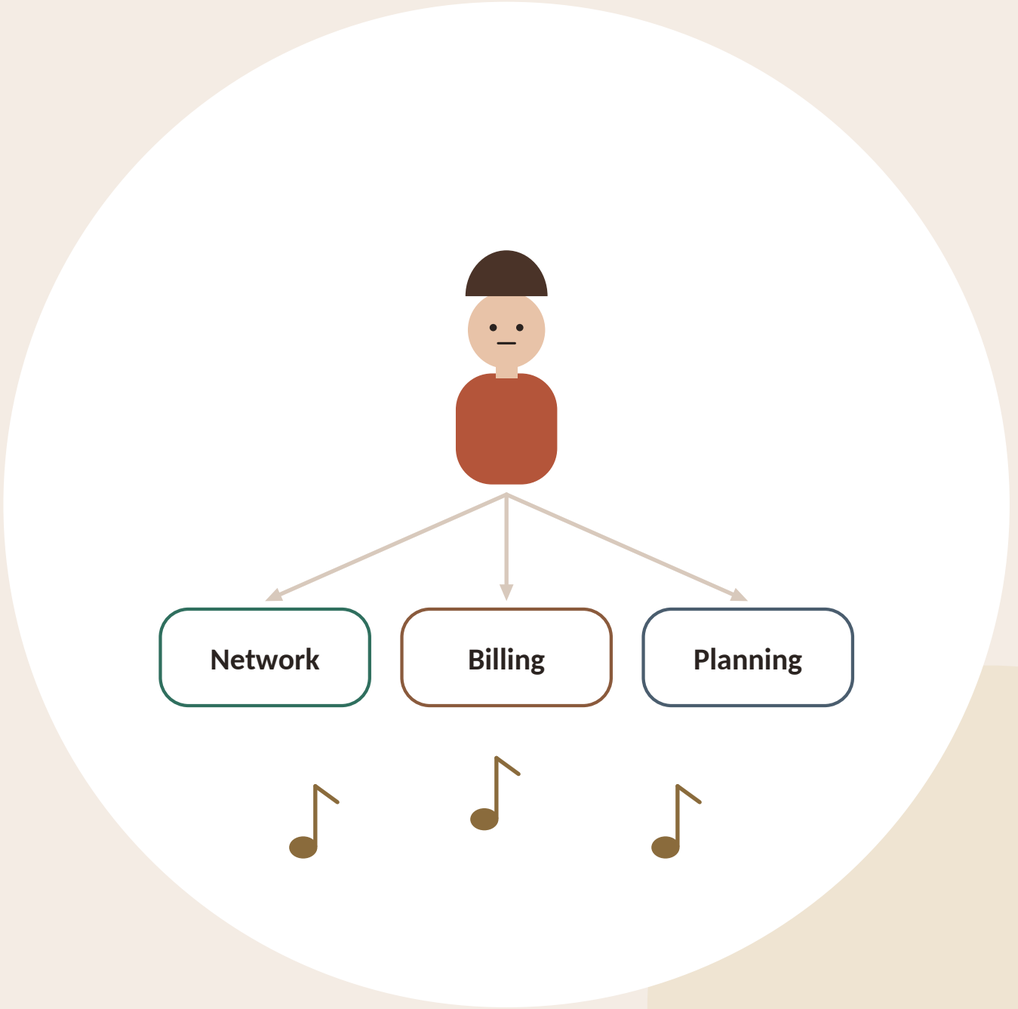
Orchestration
or choreography?

Two ways to run a process across systems.



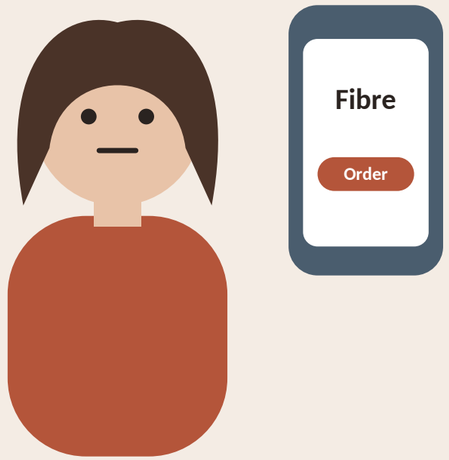
Meriem Trabelsi

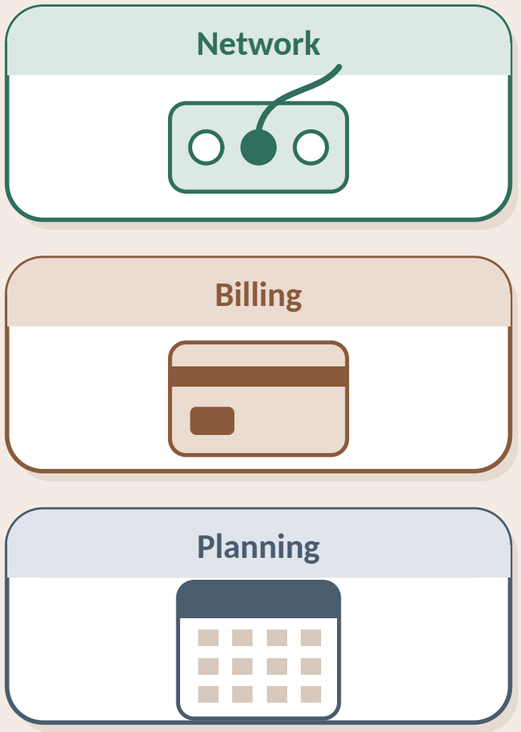
Software engineer

Who decides what happens next?

back to Sara

Last week a saga undid Sara's fibre order. But someone had to tell each system what to do, and when.





reserve a port

take the payment

book a technician

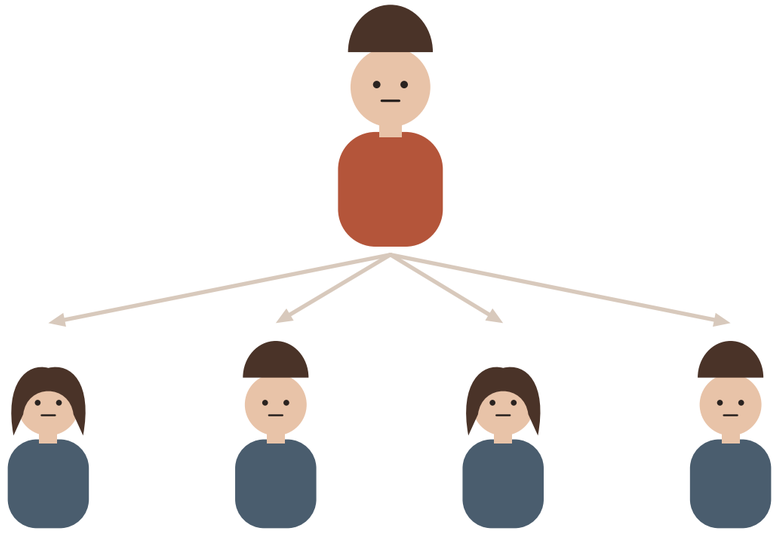
?

Two ways to coordinate

an analogy

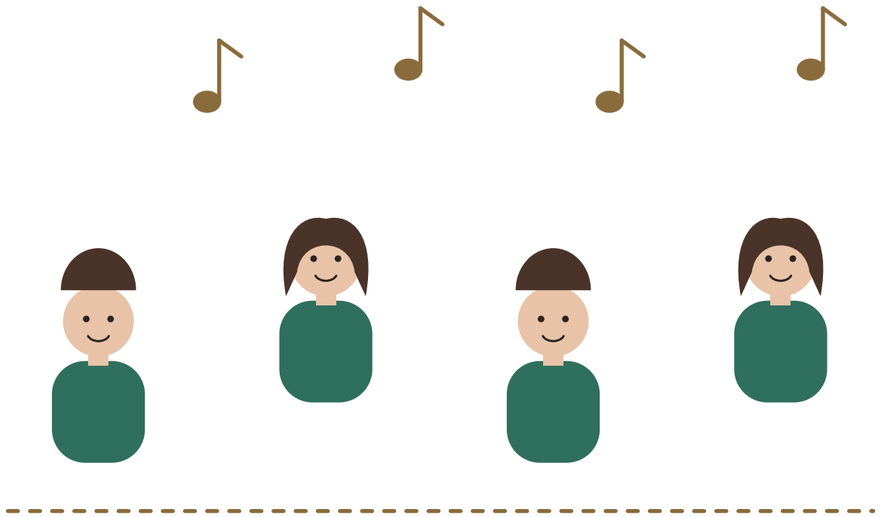
People solved this long before software. Think of an orchestra and of a dance.

An orchestra



A conductor gives every cue.

A dance

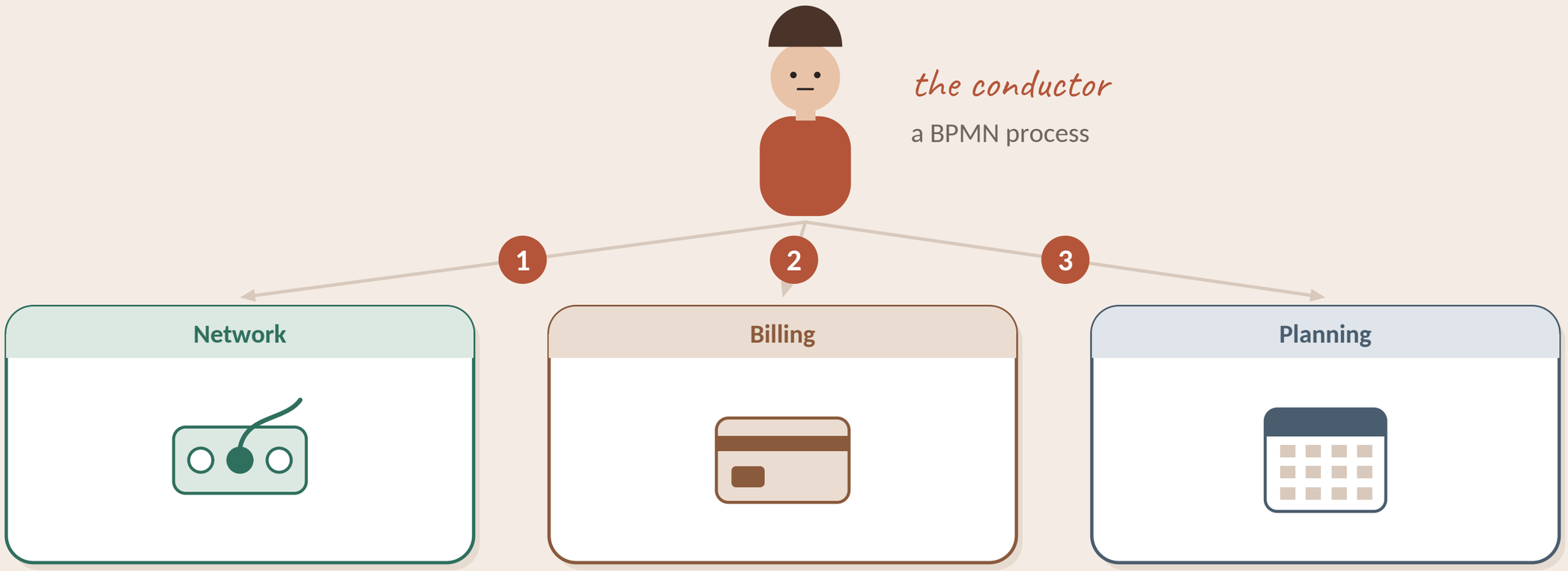


Each dancer reacts to the music.

Answer 1: orchestration

one conductor

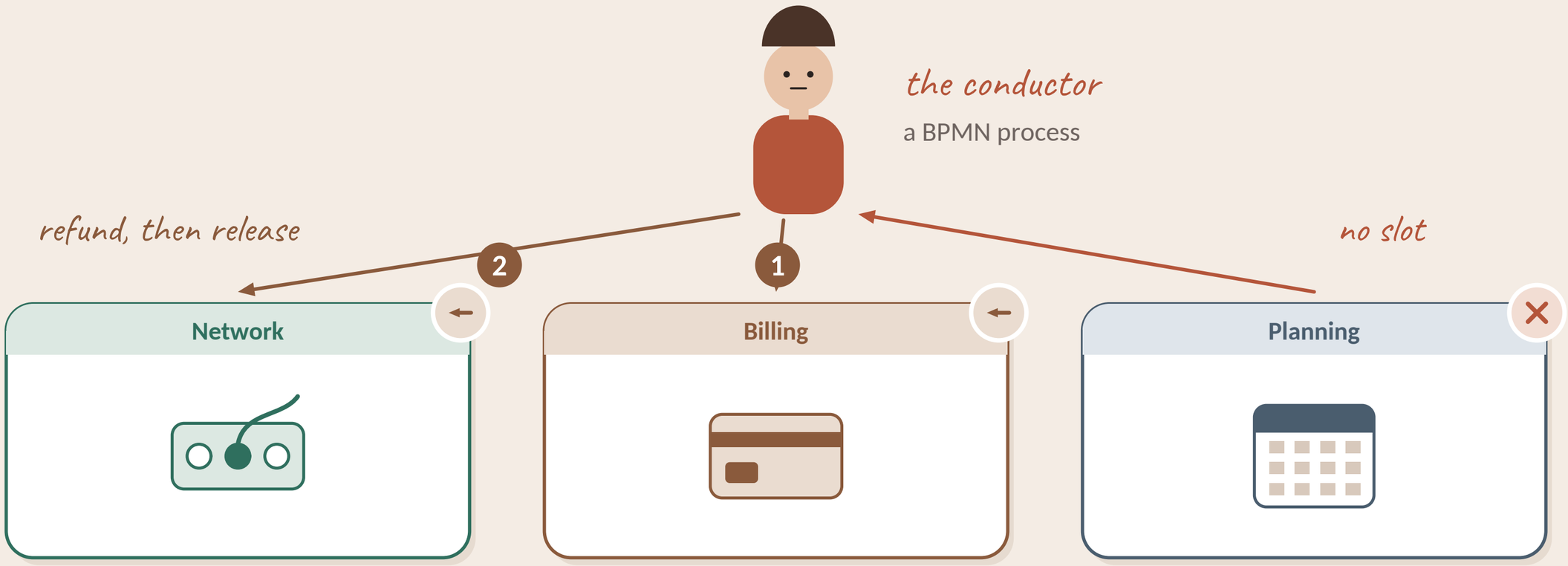
One component knows the whole order journey. It calls each system in turn and waits for the answer.



When Planning fails

orchestration

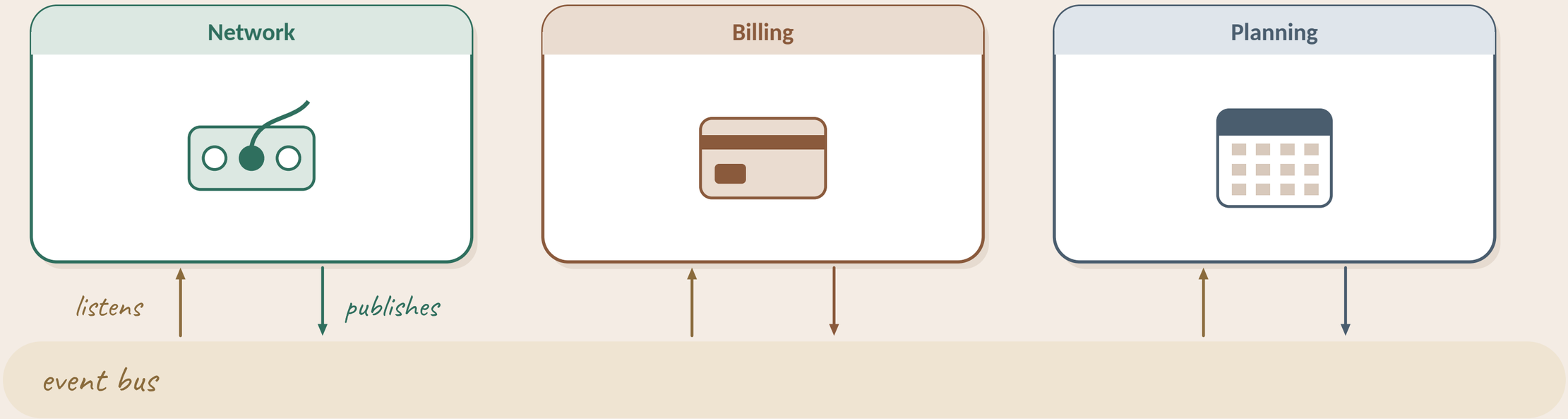
Planning answers the conductor: no slot. The conductor runs the undo steps itself, in reverse order.



Answer 2: choreography

no conductor

No one is in charge. Each system listens to an event bus, does its job, then publishes what it did.



`hears OrderPlaced`
`says PortReserved`

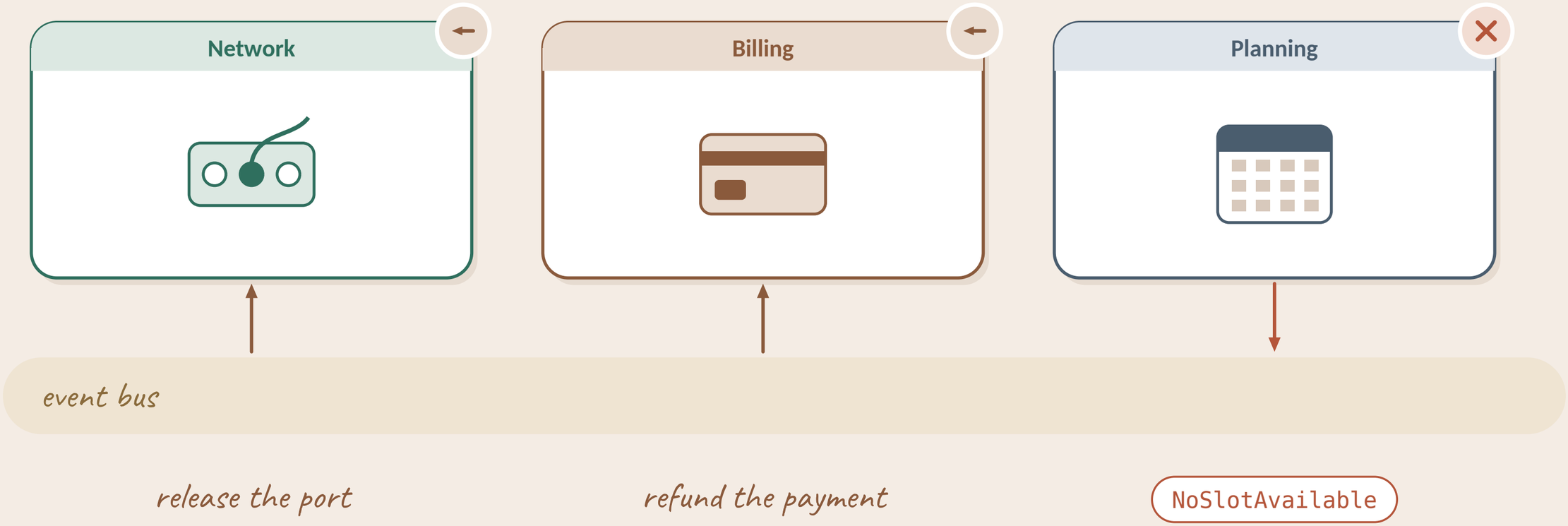
`hears PortReserved`
`says PaymentTaken`

`hears PaymentTaken`
`says SlotBooked`

When Planning fails

choreography

Planning publishes NoSlotAvailable. Network and Billing hear it and each undo their own part.



Side by side

the trade-offs

Both reach the same result for Sara. They differ in who holds the knowledge of the flow.

	Orchestration	Choreography
See the whole flow	in one place	spread across systems
Add a new step	change the conductor	add a listener
Systems depend on	the conductor	events only
Find a failure	look at the conductor	follow the events
Main risk	the conductor grows too big	nobody sees the whole picture

How to choose

one question

Start from the business, not from the technology.

Must someone see and control the whole flow?

yes

no

Orchestration

Long journeys with deadlines, compensation and audit.

Choreography

Independent teams and simple reactions like notifications.

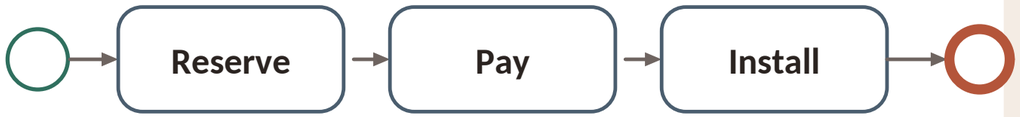
In my project I used both

my final-year project

Most real systems mix the two. Mine did: one for the journey, one for the notifications.

Orchestration

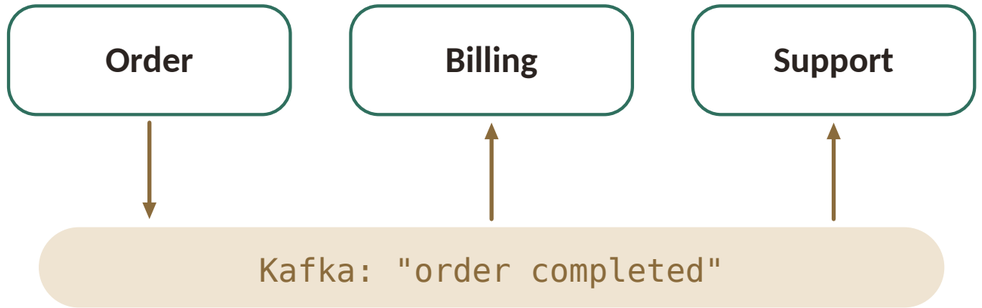
for the order journey



One BPMN process runs the steps,
the deadlines and the compensation.

Choreography

for the notifications



One service publishes. The others react
without calling each other.

Your turn

Conductor or dancers?

Which one does your team use, and why? Tell me in the comments.

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